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Brief Report

MLH1 Mismatch Repair Deficiency Predicts Exceptional Response to Immune Checkpoint Inhibition in Clear-cell Renal Cell Carcinoma

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Abstract

Immune checkpoint inhibition can result in deep and durable responses in clear-cell renal cell carcinoma (ccRCC), yet no biomarkers currently guide management. While mismatch repair (MMR) deficiency is an established predictive biomarker for immunotherapy response in other malignancies, this has not been evaluated in ccRCC, despite the proximity of MMR gene *MLH1* to *VHL* on chromosome 3p. Through genomic and immunohistochemical analyses across multiple cohorts, we identify somatic MLH1 deficiency as a relatively rare but robust predictor of exceptional immunotherapy response in ccRCC.

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ADVANCING PRACTICE

What does this study add?

Our study shows that MLH1 deficiency in clear-cell renal cell carcinoma (ccRCC) can be identified using genomic data from clinically available sequencing panels. Although rare, MLH1 deficiency may be among the most reliable predictors of an exceptional response to immunotherapy reported in ccRCC to date, an important advance in a disease with no established biomarkers to guide management.

Clinical Relevance

Defects in DNA mismatch repair (MMR) genes are well-established as predictive biomarkers of immune checkpoint inhibitor (ICI) efficacy across multiple tumour types. In metastatic clear-cell renal cell carcinoma (ccRCC), ICIs currently form the backbone of systemic therapy—both as monotherapy and in combinations with targeted agents. In this study by Dawidek et al., the authors show that loss-of-function (LoF) alterations in MMR genes are uncommon in metastatic ccRCC (~5% of cases). Notably, they identify LoF in MLH1 (2.3% of the cohort) as a potential predictive biomarker of response to immunotherapy (versus targeted therapy) in this setting, validated in two independent genomically annotated clinical-trial populations. The authors further demonstrate that MLH1 deficiency can be detected via clinically available sequencing panels, underscoring its translational feasibility to real world patient care. In contemporary practice, these findings lay the groundwork for future precision-medicine approaches in metastatic ccRCC—specifically, incorporating MLH1-status into treatment-selection algorithms to guide ICI versus targeted-agent choice. Associate Editor: Sarah P. Psutka, MD.

Patient Summary

We studied a genetic change known as MLH1 deficiency in patients with kidney cancer. We found that patients with this change responded especially well to immunotherapy and often had long-lasting benefit. This suggests that testing for MLH1 could help identify people most likely to benefit from these treatments.

In patients with metastatic clear-cell renal cell carcinoma (ccRCC), immune checkpoint inhibitor (ICI) combination therapy outperforms vascular endothelial growth factor (VEGF) targeted monotherapy [1–3]. Doublet regimens combining programmed death 1 (PD-1) and either cytotoxic T-lymphocyte–associated antigen-4 (CTLA-4) or VEGF inhibition are established first-line therapies. Nonetheless, questions persist regarding optimal patient and regimen selection, with no biomarkers to guide management [4–6].

Mismatch repair (MMR), microsatellite instability (MSI), and tumor mutational burden (TMB) are approved as tumor-agnostic biomarkers for ICIs [7]. However, TMB has not demonstrated predictive value for ICI response in ccRCC [4–6]. A small but nontrivial portion of ccRCC cases harbor MMR gene alterations, but their relationship with ICIs remains unknown [8]. We investigated whether MMR deficiency predicts an exceptional ICI response in ccRCC, with implications for both metastatic and high-risk localized disease management, especially given the recent evidence for adjuvant pembrolizumab [9].

First, we reviewed 893 patients with ccRCC sequenced using the IMPACT panel in the Memorial Sloan Kettering Cancer Center (MSK) clinical sequencing cohort, to establish the prevalence of MMR gene alterations (Fig. 1A). Among the 474 patients sequenced before receiving systemic therapy, 24 (5.1%) had loss-of-function MMR alterations. Five additional cases (1.1%) had wild-type MMR but were either MSI-high (MSIsensor >10) or TMB-high (>10 mutations/Mb). These were investigated due to potential epigenetic silencing, which causes MMR deficiency in other malignancies, but could not be detected with the available data [10].

Next, we evaluated which genomic alterations manifested as MMR protein deficiency. Specimens were available for immunohistochemical (IHC) staining in 16 of 29 cases (Fig. 1A). Only *MLH1* had consistent congruence between gene alteration and protein deficiency in the IHC-tested subset (all four of the four available cases). Among available cases with wild-type MMR but TMB-high, two of three cases were MLH1 deficient by IHC staining, supporting epigenetic silencing and suggesting that the prevalence of MMR deficiency may be underestimated. Reinforcing these findings, MMR-dominant mutational signatures (>50% attributable to MMR-associated signatures) were extracted in eight of 11 cases with genomic or IHC evidence of MLH1 deficiency, but none with the other MMR gene alterations (Supplementary Table 1).

Notably, *MLH1* is located at 3p22, between *VHL* at 3p25 and the chromatin modifying genes (*PBRM1*, *SET2*, and *BAP1*) at 3p21. Since 3p loss is the initiating event in ccRCC tumorigenesis, ccRCC should exhibit near-ubiquitous loss of heterozygosity in *MLH1*, increasing susceptibility to biallelic inactivation [11]. Indeed, allele-specific copy number analysis confirmed a high prevalence of *MLH1* loss of heterozygosity (21/29) and biallelic events (six of nine), whereas this was not the case for the other MMR genes (Supplementary Table 1). Overall, our results demonstrated that *MLH1* is the MMR alteration that is most common as well as most reliably predicts protein deficiency in ccRCC, so we focused a subsequent analysis on *MLH1*.

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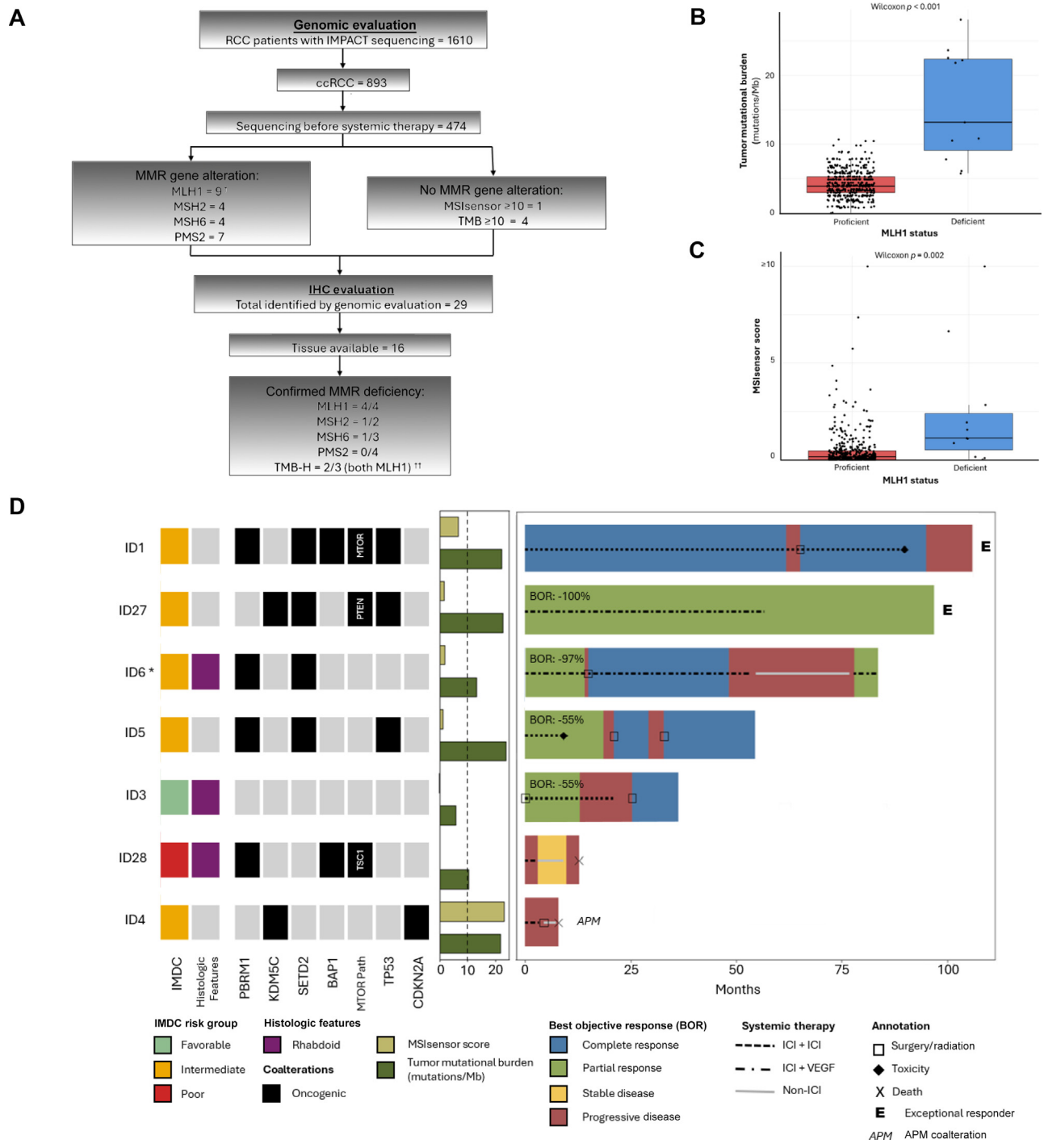


Fig. 1 – (A) MSK cohort case selection flowchart. Patients with loss-of-function alterations in MMR genes were included, as were patients with wild-type MMR genes but MSI-high or TMB-high status. IHC evaluation for MMR protein expression was performed where tissue was available. Genomic data reliably predicted protein deficiency for MLH1 only. Of the 11 patients with either genomic (†) or IHC (††) evidence for MLH1 deficiency, three were excluded for non-ICI systemic therapy and one was excluded for concurrent advanced malignancy, leaving seven cases for the final analysis. (B and C) Tumor mutational burden and MSIsensor score among the 11 cases with evidence of MLH1 deficiency compared with the remainder of metastatic ccRCC cases sequenced before systemic therapy in the MSK cohort. Boxplots represent medians with interquartile ranges. (D) Summary of genomic and clinical data for the seven MLH1-deficient cases meeting all the inclusion/exclusion criteria in the MSK cohort. Patient ID 6 (*) was notable for a germline MLH1 mutation. APM = antigen presenting machinery; ccRCC = clear-cell renal cell carcinoma; ICI = immune checkpoint inhibitor; IHC = immunohistochemistry; IMDC = International Metastatic Renal Cell Carcinoma Database Consortium; MMR = mismatch repair; MSI = microsatellite instability; MSK = Memorial Sloan Kettering Cancer Center; RCC = renal cell carcinoma; TMB = tumor mutational burden; TMB-H = TMB-high; VEGF = vascular endothelial growth factor.

Eleven patients (2.3%) had genomic or IHC evidence of MLH1 deficiency, and we evaluated the seven with metastatic ccRCC and no other advanced malignancy who received first-line ICIs (Fig. 1B and Supplementary Table 2). All received doublet PD-1/L1-based regimens (of seven, three with CTLA-4 inhibition and four with VEGF inhibi-

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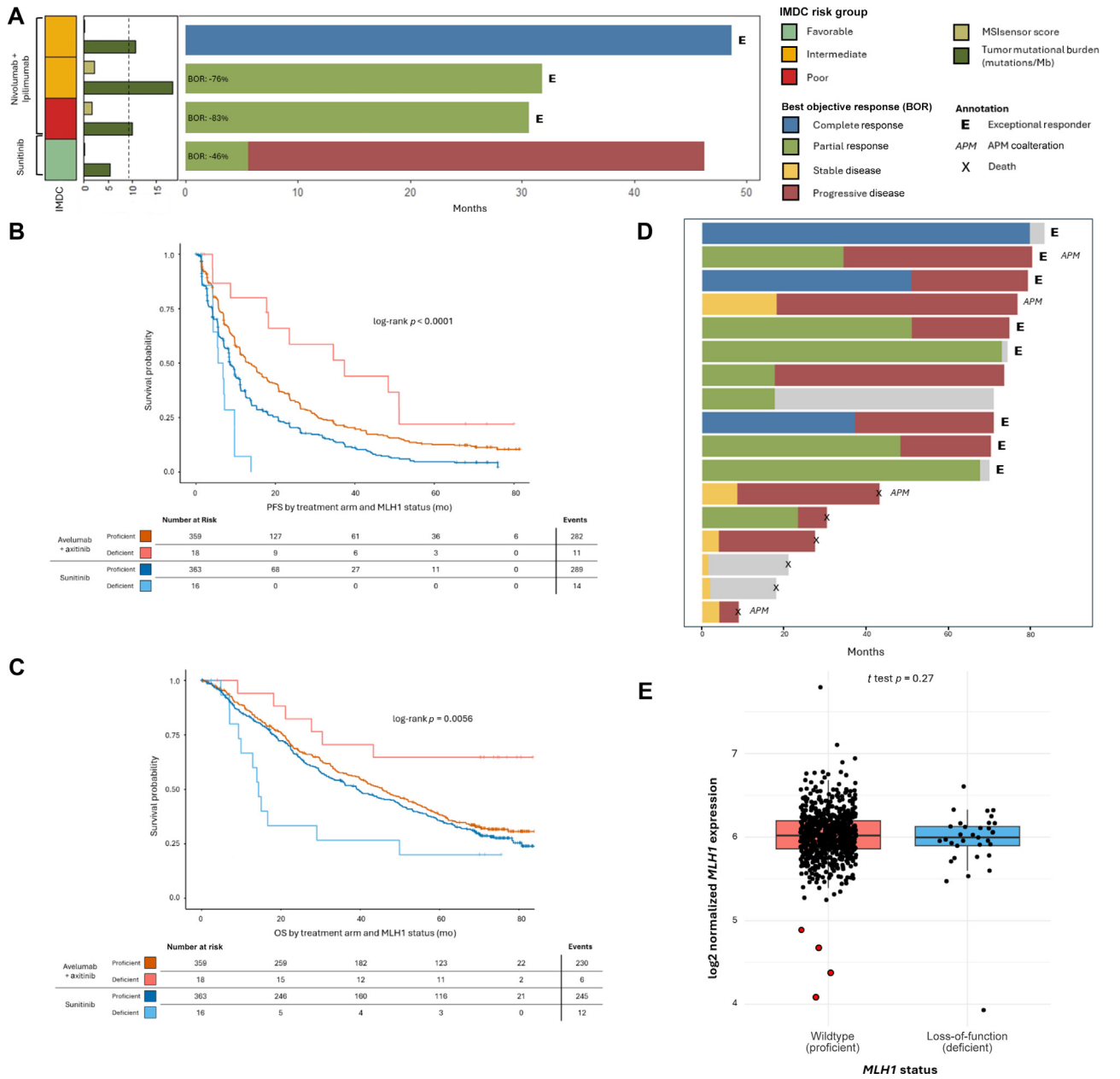


Fig. 2 – Data from two external validation cohorts. (A) Clinical course of CheckMate-214 patients with MLH1-deficient ccRCC. All patients treated with nivolumab and ipilimumab were exceptional responders, while the patient treated with sunitinib progressed rapidly. (B and C) JAVELIN Renal 101 had sufficient cases to permit a survival analysis. In the avelumab and axitinib arm, progression-free and overall survival were superior among MLH1-deficient cases, while in the sunitinib arm, outcomes were inferior among MLH1-deficient cases, suggesting that MLH1 deficiency itself is a poor prognostic factor. (D) Clinical course of JAVELIN Renal 101 patients with MLH1-deficient ccRCC treated in the avelumab and axitinib arm. (E) *MLH1* RNA expression in JAVELIN Renal 101. The cases highlighted in red had wild-type *MLH1* but low gene expression. All four were treated in the avelumab and axitinib arm, and three were exceptional responders, suggesting potential epigenetic silencing (Supplementary Table 2). Boxplots represent medians with interquartile ranges. APM = antigen presenting machinery; BOR = best objective response; ccRCC = clear-cell renal cell carcinoma; IMDC = International Metastatic Renal Cell Carcinoma Database Consortium; MSI = microsatellite instability; OS = overall survival; PFS = progression-free survival.

tion). Five achieved objective radiographic responses, with target lesions shrinking 55–100% and progression-free survival (PFS) of 13–97 mo. Two met the recently proposed Dana-Farber criteria for an exceptional response to an ICI in ccRCC (Supplementary Fig. 2) [12]. Most responders underwent local therapy for subsequent recurrences, but only one received additional systemic therapy (VEGF-targeted second-line therapy followed by successful ICI rechallenge), and all five patients remained alive after 36–106 mo of follow-up.

The hypothesized mechanism of ICI sensitivity in MMR deficiency is increased neoantigen burden [7,12]. While ccRCC generally has a low TMB (median 3.9 mutations/Mb and 2.1% TMB-high across the MSK cohort), most MLH1-deficient cases were TMB-high, with a median of 13.2 mutations/Mb (Fig. 1B). While TMB alone is not predictive of an

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ICI response in ccRCC, our findings suggest that MLH1 deficiency might impact both the quantity and the quality of mutations, enriching for immunogenic neoantigens, leading to sustained responses [4–6,12]. Interestingly, MMR-deficient mutational signatures were present despite MSIsensor scores indicating microsatellite stability (Fig. 1C). Incongruence between MMR and microsatellite status has been reported in other malignancies, but its significance remains unknown [13].

Our data provide some insight into ICI failure despite MLH1 deficiency. One exceptional responder (ID1) developed a late tumor in the contralateral kidney. Partial nephrectomy and tumor sequencing indicated an MLH1-proficient metastasis from the original primary (Supplementary Table 3 and Supplementary Fig. 3), supported by IHC evaluation (Supplementary Fig. 4). This suggests early clonal divergence and subsequent immune escape from tumor heterogeneity, a mechanism known to impair ICI effectiveness in other MMR-deficient malignancies [7]. Meanwhile, one nonresponder (ID4) had a loss-of-function mutation in *TAP1*. Concomitant deficiency of such key components of antigen presenting machinery (APM; Supplementary Fig. 5) could abrogate any proimmune effect from MLH1 deficiency [14].

We sought to corroborate our findings in independent cohorts, using data from two landmark phase 3 trials of first-line ICI therapy for metastatic ccRCC [1,2]. Among the 427 patients from CheckMate-214 for whom genomic data were available, four (0.9%) harbored *MLH1* loss-of-function alterations and were suspected to be MLH1 deficient. Three received nivolumab and ipilimumab, all meeting the criteria for an exceptional response, with target lesions shrinking 76–100% and no progression at 30.6–48.7 mo of follow-up (Fig. 2A and Supplementary Table 2). The fourth received sunitinib and achieved a partial response with rapid progression.

Among the 756 patients from JAVELIN Renal 101 for whom genomic data were available, 34 (4.5%) were suspected to be MLH1 deficient. Eighteen received avelumab and axitinib, with an 18-mo PFS rate of 73.3% (95% confidence interval [CI] 54.0–99.5%) and a 5-yr OS rate of 64.7% (95% CI 45.5–91.9%), after a median follow-up of 74.1 mo among survivors (Fig. 2B and 2C). Notably, eight of 18 (44%) met the criteria for an exceptional response, while three of six with a limited response had coalterations in APM genes (Fig. 2D). In contrast, MLH1-deficient patients who received sunitinib had inferior outcomes with no exceptional responders (Fig. 2B and 2C). The interaction term between the treatment arm and MLH1 status was significant on multivariable regression (hazard ratio 3.71, 95% CI 1.65–8.34, $p = 0.001$), suggesting that MLH1 deficiency itself is a poor prognostic feature (Supplementary Table 4).

Surprisingly, *MLH1* gene alterations did not correlate with RNA expression in the JAVELIN Renal 101 cohort (Fig. 2E). This could be attributable to mechanisms of MMR deficiency: epigenetic silencing correlates more strongly with RNA expression, whereas gene mutation causes end-protein loss of function. Interestingly, three of four patients with wild-type *MLH1* but low RNA expression achieved exceptional responses to avelumab and axitinib, again supporting epigenetic silencing and underestimated prevalence (Supplementary Table 2).

While genomic sequencing is not currently the standard practice in ccRCC management due to a limited predictive value, this could evolve as our armamentarium of therapies and understanding of outlier cases broaden. Although IHC remains the benchmark, our study demonstrates that MLH1 deficiency can be identified in ccRCC using genomic data from clinically available sequencing panels. Albeit rare, MLH1 deficiency may represent one of the most reliable predictors of an exceptional response to ICI-based therapy reported for ccRCC to date. While all metastatic patients should receive ICI-based therapy when feasible, future work could explore the implications of MLH1 status for regimen selection, such as combination with CTLA-4 versus VEGF inhibition. Furthermore, while KeyNote-564 marked a substantial breakthrough in the adjuvant space, MLH1 status could be considered in the ongoing efforts to refine patient selection.

Author contributions: A. Ari Hakimi had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Dawidek, Calderon, Coleman, Russo, Kotecha, Carlo, Reznik, Motzer, Voss, Hakimi.
Acquisition of data: Dawidek, Shah, Bakouny, Meli, Saad, Saliby, Eid, Labaki, Machaalani.
Analysis and interpretation of data: Dawidek, Calderon, Shah, Hakimi.
Drafting of the manuscript: Dawidek, Voss, Hakimi.
Critical revision of the manuscript for important intellectual content: Calderon, Khandwala, Coleman, Russo, Kotecha, Carlo, Braun, Choueiri, Motzer.
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Supplementary data

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